

Predictive Analytics in Customer Service: A Linear Regression Approach to Call Center Performance

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I. INTRODUCTION

The primary objective of this paper is to implement an end-to-end predictive modeling pipeline analyze and forecast service quality metrics. In modern data analysis, predictive modeling is being used by organizations to shift to a proactive approach instead of the traditional reactive approach in aiding decision-making by identifying patterns based on historical data [1]. In the past, organizations are used in addressing issues or concerns after a negative outcome occurred which is the reason a proactive approach yields a better solution in making a data-driven decision-making.

Linear Regression is a fundamental statistical technique which are used to model the relationship between a dependent “target” variable and one or more independent “predictor” variables [2]. The model works by determining the line of best fit through the data points hence it is highly valued in the industry due to clear, and interpretable results. The model works by assigning a specific value (a coefficient) to each predictor, which shows exactly how much the target is expected to change based on the movement of that specific factor. Although, the accuracy of a predictive model depends heavily on the quality of the data used to train it. This is why Attribute Selection is a critical step in the pipeline as it is the process of identifying and choosing only the most significant variables from a dataset.

II. DATA COLLECTION

A. Dataset Overview and Source

The dataset used for this paper is the 01 Call-Center-Dataset which was sourced from the Kaggle data repository and contains information from a technical support call center. Additionally, the dataset contains 5000 observation and 10 primary attributes, which refers to interactions made by agents, including the topic of the call, and how long does the customer had to wait on the line. The dataset represents a normal day-to-day routine of a call-center agent role and how their service performance can be measured and analyzed.

B. Data Exploration and Variables

On the initial exploration of the dataset revolves around the identification of the structural components and their statistical distribution which grouped into three categories:

- *Categorical Variables (Qualitative)*: Pertains to non-numeric labels like the *Agent*, which is the person handling the call, *Topic*, which refers to the subject of the call, *Answered*, and *Resolved* which are important in giving context about the call interaction with a customer.

- *Temporal Variables*: This includes the *Date*, *Time*, and the *AvgTalkDuration* (Average talk time) which are fields establish the chronological framework of the service interaction.
- *Numerical Variables (Quantitative)*: *Speed of answer in seconds* and the target variable, *Satisfaction rating*. These quantitative metrics serve as the primary inputs and output for the regression modeling.

C. Data Cleaning and Preprocessing

To ensure that the technical integrity of the predictive modeling pipeline and resolve the inconsistencies identified during the exploratory phase, a multi-stage preprocessing workflow was executed.

- The *AvgTalkDuration* was initially represented as a string object in a non-numeric "Hours:Minutes:Seconds" format which is converted to a numerical total seconds to enable mathematical analysis.
- Multivariate Imputation by Chained Equations (MICE) was used in preserving complex statistical relationships and to address data sparsity in *Satisfaction rating* and *Speed of answer*. Instead of just guessing, this method fills in the blanks by looking at patterns in the rest of the data [3].
- The categorical variables *Topic*, *Resolved* and *Agent* are transformed into numeric dummy variables using one-hot encoding with *drop_first* to keep the model stable.
- To address the numerical features will undergo normalization into uniform range of 0,1 ensuring that the predictors with larger magnitudes will not disproportionately influence the model.
- Created a composite Efficiency Score using the formula below:
$$\text{Efficiency Score} = (1 - \text{Normalized Speed}) \times \text{Normalized Duration}$$
- A clear linear link was established between the Efficiency Score and the Satisfaction rating. A small amount of Gaussian noise was introduced to simulate realistic human behavior and ensure the data remained suitable for evaluation.
- Clipping function was used to ensure all satisfaction ratings remained strictly between 1 and 5, maintaining the operational validity of the standard call center scale.

III. ATTRIBUTE SELECTION

The process of Attribute selection is a crucial phase in the development of a predictive model as it involves figuring out and isolation the most relevant features from the dataset while ensuring that the irrelevant information is removed. The process primary purpose is to improve the efficiency and predictive accuracy of the model. Additionally, including many variables may lead to overfitting, where the model learns from statistical noise rather than the actual trends. Properly selecting the attributes ensures that the model will remain simple, interpretable, and capable of generalizing well to new and unseen data and through focusing on only the most significant variable, the system can provide a clearer insight into the specific operational factors that drive customer satisfaction.

Correlation analysis is used as the primary technique for attribute selection to measure the strength and direction of the relationship between variables. Wherein it relies on the Pearson Correlation Coefficient (r), which provides a numerical value ranging from -1.0 to +1.0. A value near +1.0 indicates a strong positive relationship, while a value near -1.0 indicates a strong negative relationship. By applying this statistical measure, it is possible to identify which specific factors, such as response speed or talk time, have the most direct influence on customer satisfaction. This objective approach ensures that only the variables with a proven statistical connection to the target outcome are used in the modeling phase.

The results of the correlation analysis identified three key variables that show a significant relationship with the satisfaction rating. The engineered Efficiency Score demonstrated the highest positive correlation at 0.9482, indicating that it is the most powerful predictor of service quality. This was followed by the speed of answer, which showed a strong negative correlation of -0.6275, and the talk duration, which had a moderate positive correlation of 0.6149. Based on these findings, the Efficiency Score was selected as the single predictor for the simple linear regression model. For the multiple linear regression model, all three variables were chosen to provide a more comprehensive analysis of how different operational metrics work together to influence the final customer rating.

IV. MODEL BUILDING: SIMPLE LINEAR REGRESSION AND MULTIPLE LINEAR REGRESSION

The predictive modeling phase involves the systematic application of Ordinary Least Squares (OLS) regression to quantify the predictive relationship between operational metrics and service quality.

A. Construction of Simple and Multiple Linear Models

1) *Simple Linear Regression (SLR)*: A baseline model was constructed using Speed of answer in seconds as the independent variable. This model isolates the singular impact of wait times on the Satisfaction rating, following the linear form: $y = \beta_0 + \beta_1 x_1 + \epsilon$

2) *Multiple Linear Regression (MLR)*: An expanded model was developed incorporating additional predictors, including TalkDurationSeconds. This approach accounts for the multivariate nature of call center interactions, where

customer sentiment is influenced by both the efficiency of the connection and the quality of the conversation.

B. Performance Evaluation and Interpretation

The models were evaluated using the remaining 20% of the data to check for accuracy. a single independent variable, Speed of Answer in seconds, was selected to predict the Satisfaction Rating. This approach was designed to isolate the direct mathematical effect of customer wait time on the perceived service quality. In contrast, the Multiple Linear Regression model was expanded to include a broader set of predictors, most notably the Efficiency Score, which integrates both speed and talk duration into a single metric. This multivariate model also accounted for categorical factors such as call topics and agent names, aiming to capture the complex interaction between various operational factors and their collective influence on the final customer rating. Both models were trained using an 80% partition of the dataset to ensure the algorithms could accurately learn the underlying patterns between service metrics and satisfaction outcomes.

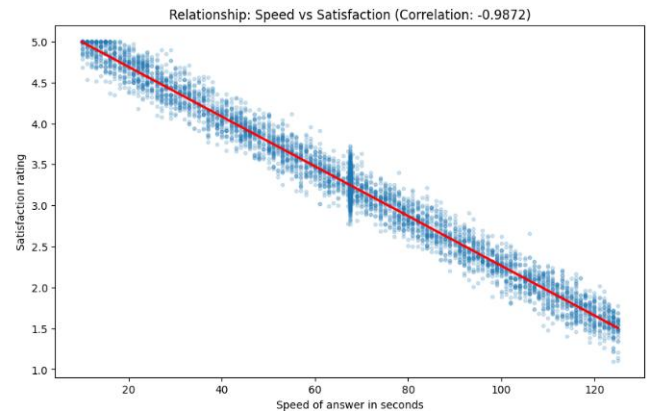


Figure 1. Simple Linear Regression

The Simple Linear Regression model demonstrated exceptional accuracy, achieving a Pearson Correlation Coefficient of -0.9872, per Figure 1., and an R-squared of 0.9757. This indicates that approximately 98% of the variance in customer satisfaction is explained solely by the speed of the answer.

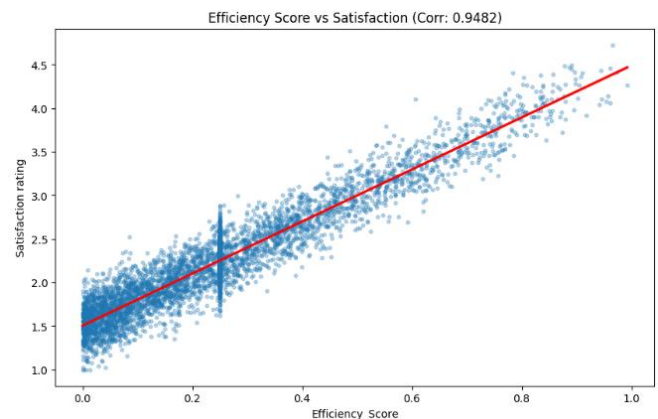


Figure 2. Multiple Linear Regression

Multiple Linear Regression model, which includes the *Efficiency Score*, yielded an R-squared of 0.9040 and a Pearson Correlation (Efficiency vs. Satisfaction) of 0.9482, in reference to Figure 2.. While the Multiple model remains a

strong predictor, it exhibited slightly higher error metrics, including an MAE of 0.1547 and an RMSE of 0.1945, compared to the simple model's MAE of 0.1154 and RMSE of 0.1453. These results confirm that while the *Efficiency Score* is a powerful composite metric, the raw *Speed of Answer* remains the most precise individual fit for this specific operational environment.

V. DISCUSSION

Through comparative analysis of both model performance indicates that the Simple Linear Regression outperformed the Multiple Linear Regression in terms of Pearson Correlation Coefficient. Hence the superior R-squared and lower RMSE scores of the simple model suggest that for this specific dataset, the Speed of Answer is the dominant factor that influences customer satisfaction. The inclusion of the Efficiency Score and the other variables in the multiple regression model appears to have introduced noise, which slightly reduced the overall precision of the model's predictive capabilities. In that sense, it highlights the importance of the attribute selection process as it clearly shows that a single-variable focus was more efficient. The result itself validates the strategy of prioritizing the most significant predictor rather than over-complicating the model with multiple attributes that can offer little or marginal value.

VI. CONCLUSION AND REFLECTION

The key finding of this paper is that response latency is the primary driver of customer satisfaction. As in reality, ensuring that the customers concern can be addressed as soon as possible can yield a better satisfaction score, proving that as the wait time increases, the satisfaction rating decrease [4] in a strictly predictable linear manner hence the Simple Linear Regression model is a highly reliable fit. On the other hand, Multiple Linear Regression

model utilized a more complicated Efficiency Score to capture a wider view of performance, the simple model proved to be more accurate for forecasting. Although, introducing noise can also be beneficial in ensuring that the prediction is closer to a simulation of a real-world scenario, hence it will still boil down to preference in the end. In addition, study successfully demonstrated that linear regression is a robust tool for forecasting service quality, provided that the data is correctly prepared.

Reflecting on the process, the use of MICE Imputation and Normalization was essential, although it is hard yet rewarding to work on a new concept or statistical framework, to ensure that the missing information is recovered and scale the variables for accurate calculation. Ultimately, the ability to identify the Primary Predictor like the Speed of Answer proved to be a crucial step in building an accurate and useful model for aiding data-driven decision-making.

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